

Extremal mass at fixed angular momentum in EGB, $D = 5$

Status as of 9 October 2026. Asymptotically flat black holes, two equal spins $J_1 = J_2 = J$, $G_5 = 1$, action $R + \alpha \mathcal{L}_{\text{GB}}$.

We want the smallest mass $M_{\text{ext}}(J, \alpha)$ that still has a horizon when J is held fixed and the Gauss–Bonnet coupling α is varied. By dimensional analysis it is a function of one variable:

$$M_{\text{ext}} = \pi^{1/3} J^{2/3} m(s), \quad s = \frac{\pi^{2/3} \alpha}{J^{2/3}}.$$

No closed form for $m(s)$ is known. This is what we have.

1. Weak coupling: the series through s^4 .

$$m(s) = \frac{3}{2} + s - \frac{236}{105} s^2 + d_3 s^3 + d_4 s^4 + O(s^5),$$

$$d_3 = \frac{60851752}{33075} - 1536 \zeta(3) = -6.548307\dots, \quad d_4 = -\frac{592541765354296}{1489863375} + \frac{227328}{5} \zeta(3) + \frac{88064}{25} \pi^4 = 66.100522\dots$$

The linear term is known [1]. The coefficients d_2 and d_3 were first obtained from the Euclidean action. We have now rederived them in a different way: we solve for the exterior metric of the extremal black hole order by order in α , exactly, and read the mass directly off the fall-off of g_{tt} at infinity. At each order the metric functions are combinations of multiple polylogarithms in $1/r^2$ with rational coefficients, of weight at most the order. The integration constants are fixed by asymptotic flatness and by requiring the horizon to be a double zero of g^{rr} (extremality). This is how d_4 was obtained; before, it was known only numerically. An independent numerical calculation (spectral boundary-value problem) gives 66.10052, which agrees in all seven available digits.

Checks: at each order the solution satisfies the reduced field equations exactly, and the horizon angular velocity, computed separately from the mass, satisfies the extremal first law $dM_{\text{ext}} = 2\Omega_H dJ$ (at fixed α) with zero residual at orders 2, 3 and 4.

Caveat: d_4 is exact *provided* certain constants that appear when the polylogarithms are evaluated at the horizon equal the zeta values we identified numerically (100 digits, error $< 10^{-80}$). This is not proved. The series is also formal: we do not know whether it converges. The coefficients (1, -2.2 , -6.5 , 66) do not yet show a clear growth pattern.

2. Strong coupling ($\alpha \rightarrow \infty$, J fixed): a candidate. A matched analysis of three regions (horizon throat, core, intermediate zone) gives

$$M_{\text{ext}} = \frac{3\pi\alpha}{4} + \frac{4J}{3\sqrt{\alpha}} + o(J/\sqrt{\alpha}), \quad \Omega_H \sqrt{\alpha} \rightarrow \frac{2}{3},$$

consistent with the first law. It relies on the regions matching uniformly, which is not proved. The full numerical solution reaches only $\alpha \simeq 3.12$ (with $J/\alpha^{3/2} \simeq 0.26$), too far from the limit to confirm it.

3. The point $J = 0$. The term $3\pi\alpha/4$ is the mass of the static solution with zero horizon radius, which is singular. Kleihaus, Kunz and Radu conjecture that the rotating extremal branch ends there [2]. It is not a regular extremal black hole: static black holes with a horizon always have positive temperature.

4. What is missing. There is no closed form for $m(s)$ at finite coupling, and no proof that none exists. The negative results we have apply to specific families: Hamilton–Jacobi superpotentials with finitely many powers of r , Liouvillian solutions of one perturbation equation (its differential Galois group is $SL(2, \mathbb{C})$), and a candidate differential equation for $m(s)$ that an independent numerical point rules out. The second-order exterior already contains $\text{Li}_2(1/r^2)$, and $\zeta(3)$ and π^4 appear at higher orders, which makes a simple algebraic formula unlikely.

Next steps. The s^5 term with the same program (the constants it needs are already identified); with five or six coefficients the radius of convergence can be estimated. On the numerical side, extending the branch beyond $\alpha \simeq 3$ to test the strong-coupling limit.